Diameter Growth Equations for CONUS with Environment and Climate Variables

Greg Johnson, David Marshall, Aaron Weiskittel

2026-03-30

Preliminary CONUS-wide Diameter Growth Equation

We built a data set from Forest Inventory and Analysis (FIA) remeasurement data, filtering out observations with missing values for diameter at breast height (`dbh`), basal area in larger trees (`bal`), crown ratio (`cr`), total height (`ht`), site index (`si`), elevation above sea level (`elev`), extreme minimum temperature (`EMT`), and remeasurement intervals less than 5 years. We also restricted the data to species with $\geq$ 5000 observations.

The diameter growth equation fit to these data was:

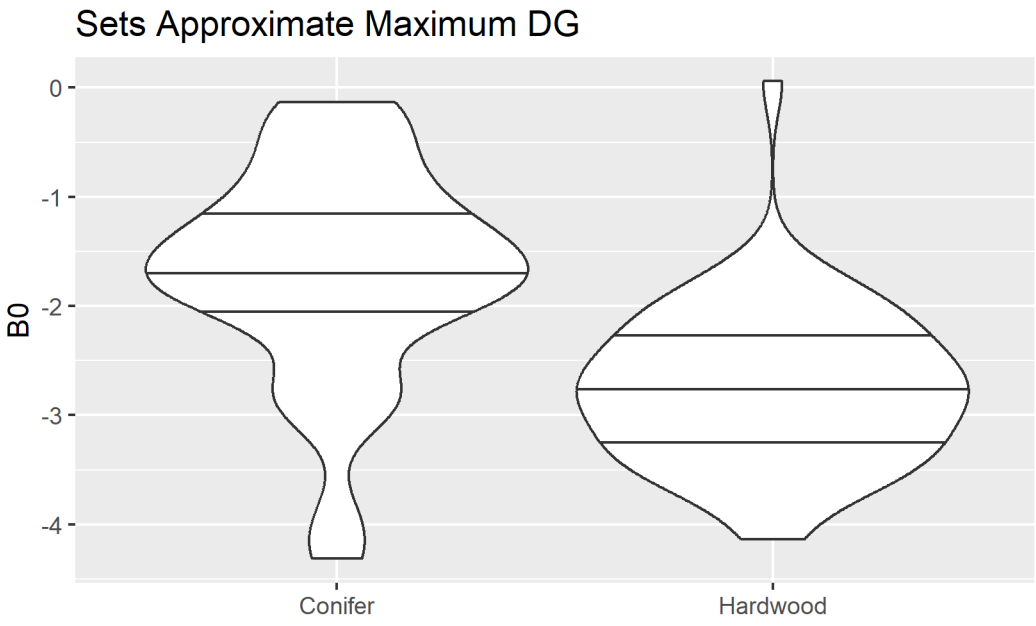
$$\Delta dbh = e^{(\beta_0 + \beta_1 log(\frac{(dbh+1)^2}{(cr*ht+1)^{\beta_3}}) + \beta_2 \frac{bal^{\beta_4}}{dbh+2.7} + \beta_5 elev + \beta_6 EMT)} \tag{1}$$

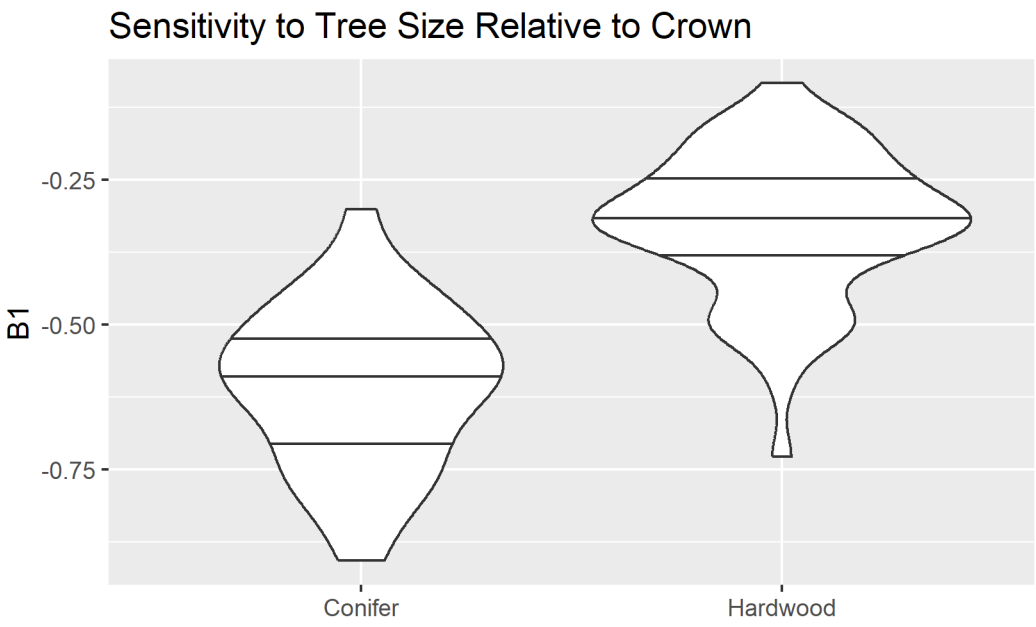
where:

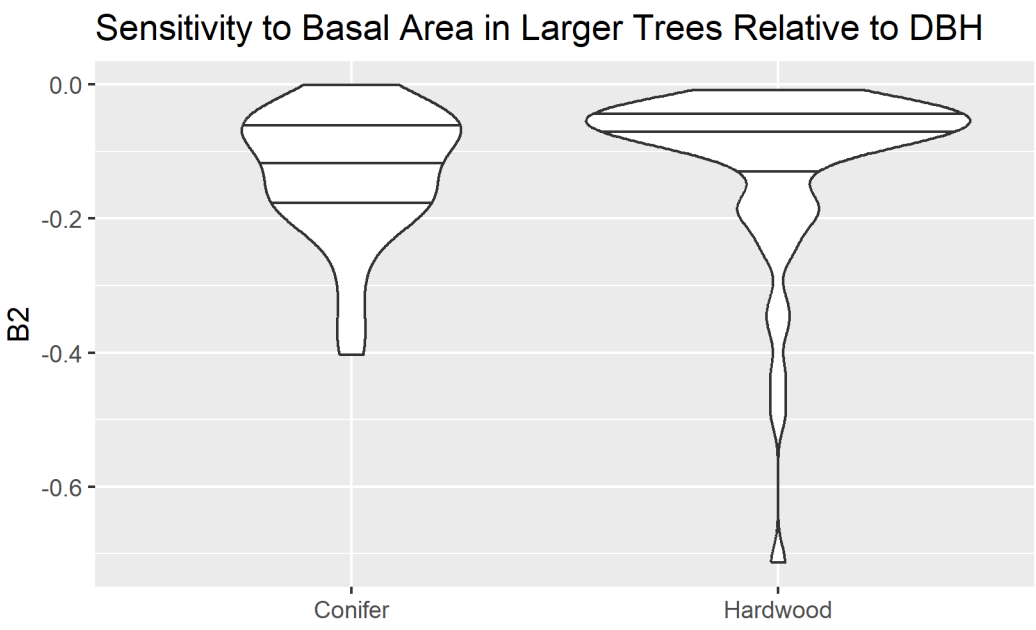
- dbh = diameter at breast height (inches)
- cr = crown ratio (fraction of total height)
- ht = total height (feet)
- bal = basal area in larger trees ($feet^2/acre$)
- $elev$ = elevation above sea level (feet)
- EMT = extreme minimum temperature ($^\circ C$)

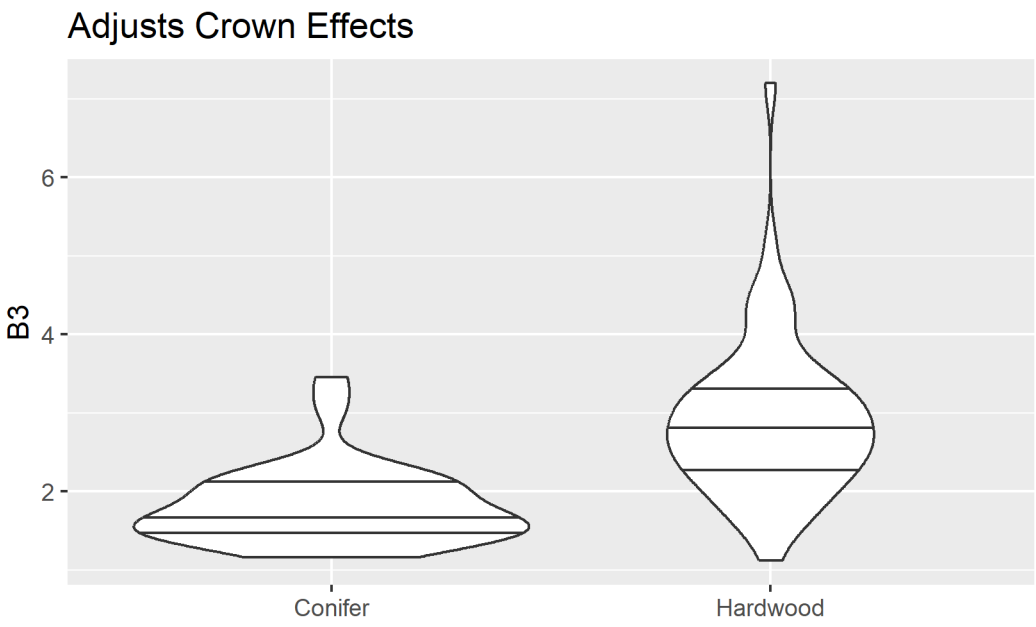
Equation 1 was fit to each qualifying species using an integrated fitting approach and errors minimized were ending `dbh`. 84 species equations were estimated. Sample sizes ranged from 5099 to 283321 observations.

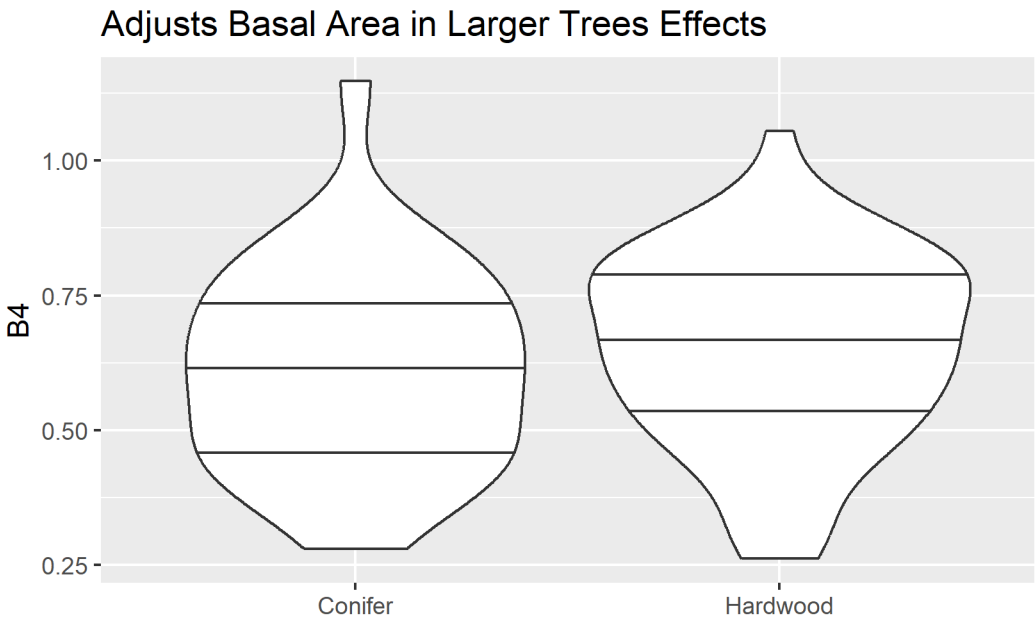
The graphs below show the range of parameter estimates divided into Conifer and Hardwood groups. The parameter estimates are remarkable stable across species.

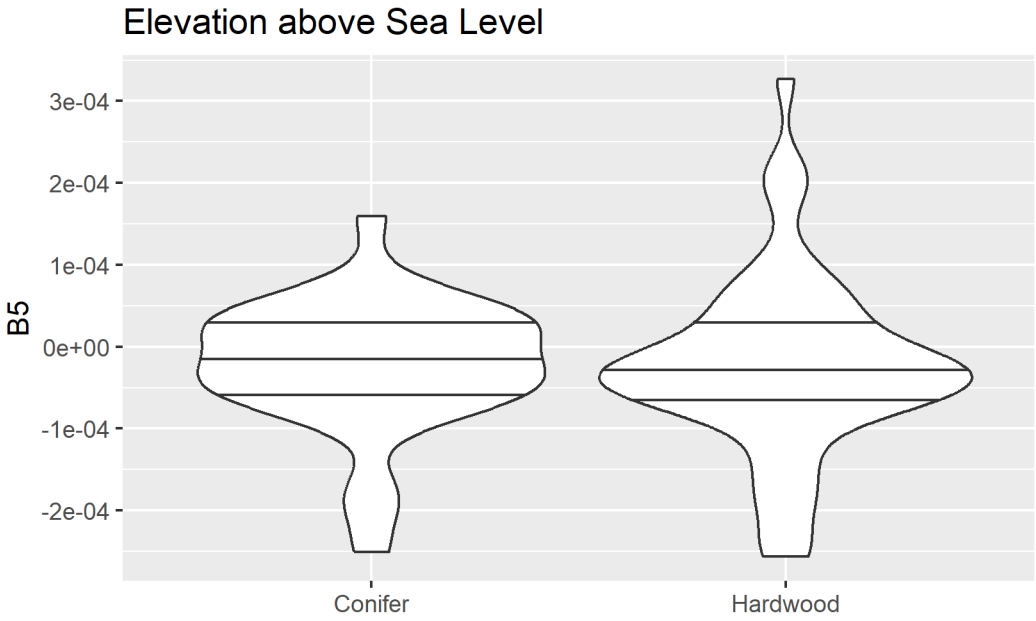

Sets Approximate Maximum DG
B0
0
-1
-2
-3
-4
Conifer
Hardwood

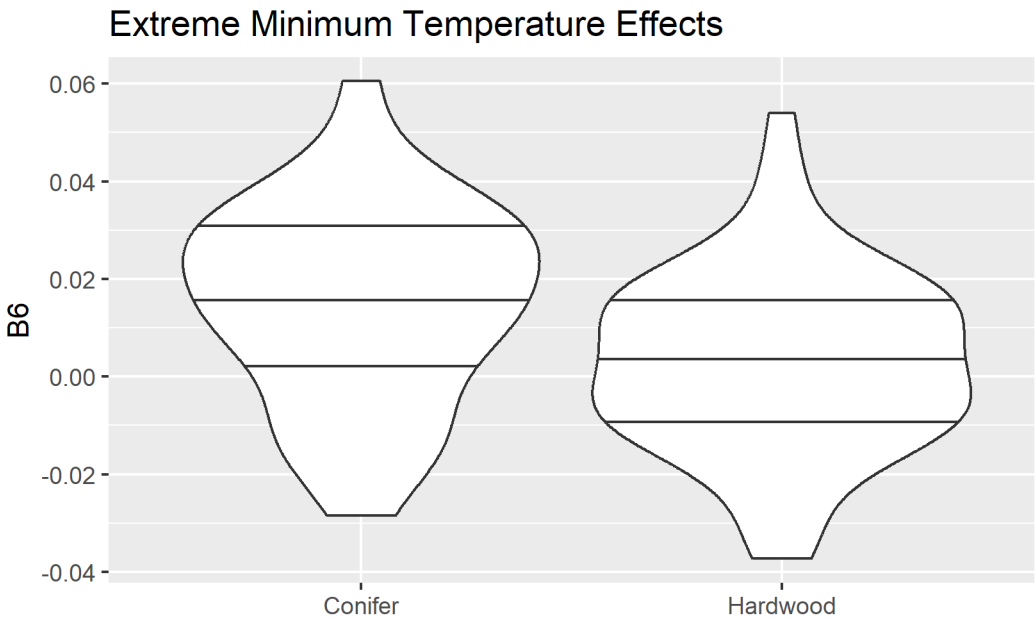

Sensitivity to Tree Size Relative to Crown
B1
-0.25
-0.50
-0.75
Conifer
Hardwood

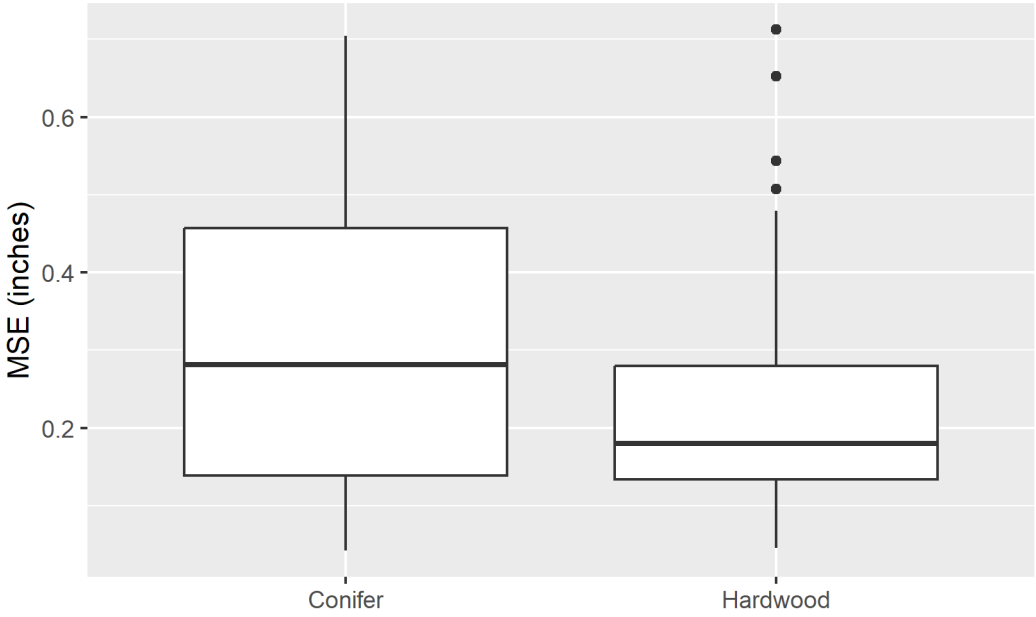

Sensitivity to Basal Area in Larger Trees Relative to DBH
B2
0.0
-0.2
-0.4
-0.6
Conifer
Hardwood


Adjusts Crown Effects
B3
6
4
2
Conifer
Hardwood


Adjusts Basal Area in Larger Trees Effects
B4
1.00
0.75
0.50
0.25
Conifer
Hardwood


Elevation above Sea Level
B5
3e-04
2e-04
1e-04
0e+00
-1e-04
-2e-04
Conifer
Hardwood


Extreme Minimum Temperature Effects
B6
0.06
0.04
0.02
0.00
-0.02
-0.04
Conifer
Hardwood


MSE (inches)
0.6
0.4
0.2
Conifer
Hardwood

FIA Species Clusters by Diameter Growth Parameter Estimates

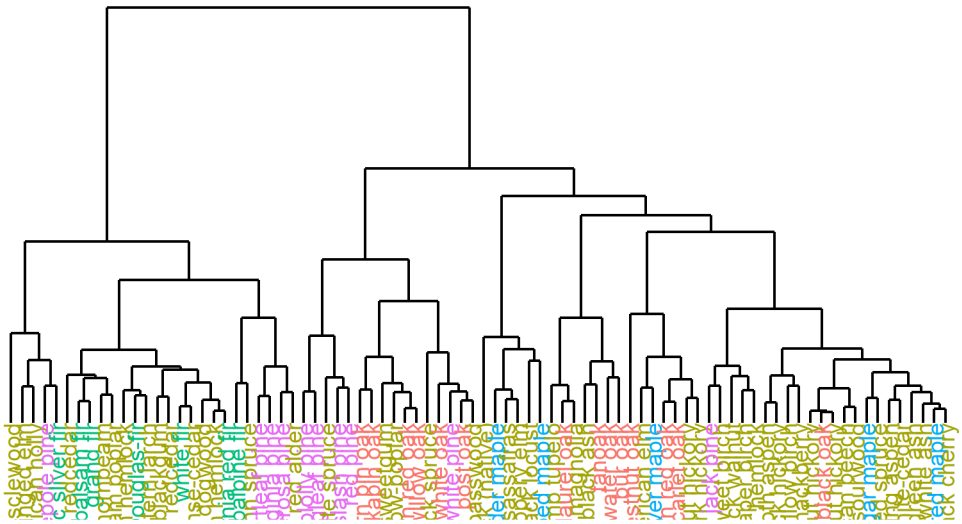


Table 1: Species with Negative Summer Precipitation Parameter

FIA Species Code	Common Name

Table 2: Species with Negative Extreme Minimum Temperature Parameter

FIA Species Code	Common Name
131	loblolly pine
611	sweetgum
746	quaking aspen
111	slash pine
621	yellow-poplar
531	American beech
375	paper birch
837	black oak
261	eastern hemlock
972	American elm
125	red pine
812	southern red oak
371	yellow birch
121	longleaf pine

FIA Species Code	Common Name
711	sourwood
409	mockernut hickory
403	pignut hickory
105	jack pine
94	white spruce
372	sweet birch
931	sassafras
313	boxelder maple
831	willow oak
602	black walnut
462	hackberry
975	slippery elm
901	black locust
408	black hickory
402	bitternut hickory
315	striped maple
826	chinkapin oak